Field application of a commercial porcine reproductive and respiratory syndrome virus (PRRSV) oral fluid antibody enzyme-linked immunosorbent assay (ELISA)

Emily Croft, Tim Blackwell, Jeff Zimmerman

Abstract — A commercial porcine reproductive and respiratory syndrome virus (PRRSV) oral fluid antibody enzyme-linked immunosorbent assay (ELISA) was used on 31 commercial swine farms in Ontario using oral fluid samples (~6 per herd) collected from cotton ropes. Using the manufacturer's cutoff [sample-to-positive ratio (S/P) ≥ 0.4], 2 of 135 oral fluid samples from 23 PRRSV presumed negative herds tested positive (1.5% false positive rate). Three approaches to improving test diagnostic specificity were compared: i) use a cutoff of S/P ≥ 0.8 for individual oral fluid samples; ii) use the current cutoff of S/P ≥ 0.4 but use a mean S/P based on several oral fluid samples (6 samples were used in this study); and iii) use serial testing to resolve unexpected positive ELISA results, i.e., retest using a reverse transcription-polymerase chain reaction (RT-PCR) to determine whether low positive S/P ratios are the result of early PRRSV infection in a barn.

Résumé — Application sur le terrain d'une épreuve immuno-enzymatique (ELISA) commerciale pour détecter des anticorps contre le virus du syndrome reproducteur et respiratoire porcin en utilisant des fluides oraux. Une épreuve immuno-enzymatique (ELISA) commerciale pour détecter des anticorps contre le virus du syndrome reproducteur et respiratoire porcin (VSRRP) en utilisant des fluides oraux fut utilisée sur 31 fermes commerciales en Ontario en utilisant des échantillons de fluides oraux (~6 par troupeau) prélevés en utilisant des cordes en coton. En utilisant le seuil recommandé par le manufacturier [ratio échantillon-à-positif (S/P) ≥ 0,4], 2 des 135 échantillons de fluides oraux provenant de 23 troupeaux présumés négatifs pour le VSRRP ont testé positif (taux de faux positifs de 1,5 %). Trois approches pour améliorer la spécificité du test furent comparées : i) utilisation d'une valeur seuil de S/P ≥ 0,8 pour les échantillons de fluides oraux individuels; ii) utilisant de la valeur seuil actuelle S/P ≥ 0,4 mais utiliser une S/P moyenne basée sur plusieurs échantillons de fluides oraux (6 échantillons furent utilisés dans la présente étude); et iii) utiliser des tests en série pour résoudre les résultats ELISA positifs non-attendus; retester en utilisant la réaction d'amplification en chaine par la polymérase avec la transcriptase reverse (RT-PCR) afin de déterminer si les ratios S/P faiblement positifs sont le résultat d'une infection débutante dans une ferme.

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Porcine reproductive and respiratory syndrome virus (PRRSV)-infected herds often experience large economic losses due to PRRSV-induced reproductive losses in sows and respiratory disease in growing pigs. Preventing and/or avoiding losses must be a prime objective of every successful pig producer, but PRRS management decisions require accurate and timely information, i.e., surveillance data.

Particularly when used in coordination with PRRSV reverse-transcription polymerase chain reaction (RT-PCR) testing, PRRSV antibody-based surveillance can provide actionable information regarding herd PRRSV status, virus circulation, and/or vaccination history (1). Traditionally, serum samples have been used in PRRSV antibody testing, but oral fluid samples likewise contain detectable levels of PRRSV-specific antibody and commercial PRRSV oral fluid ELISAs are available (2). In addition, oral fluid sampling is less invasive for pigs and more convenient for stock people. Oral fluid-based PRRSV surveillance can be particularly useful in low prevalence situations or

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when large numbers of animals need to be sampled routinely over time. Oral fluids are as diagnostically effective as sera at detecting antibody positive pigs, but unexpected positive results have raised concerns and limited the use of oral fluids on farms for surveillance purposes (3). Therefore, the objectives of this study were: i) to compare the results of PRRSV oral fluid antibody testing in Ontario swine herds to the manufacturer's stated values for test diagnostic sensitivity and specificity, and ii) to compare 3 methods to address unexpected positive test results.

To compare the accuracy of PRRSV oral fluid antibody testing in Ontario swine herds to the manufacturer's stated values, samples were collected from a convenience sample of herds (n = 31) located in southern Ontario. Herds were initially categorized as PRRSV naïve (non-infected/non-vaccinated; n = 26) or infected (n = 5) based on the producer's assessment, which may or may not have included recent laboratory testing. This approach was chosen to reflect how the test was being applied in Ontario and to determine if the unexpected positive results anecdotally reported by producers and veterinarians would be observed. A priority was given to testing PRRS naïve herds to evaluate if complaints from the field regarding false positive results were routine.

Oral fluid samples were collected by hanging lengths of cotton rope (1.6 cm diameter, ~30 cm in length) in 6 pens throughout each barn. Only 1 barn per farm was sampled. Samples were therefore clustered within farms. Six ropes per barn were used to approximate the common field application of oral fluid sampling for PRRSV for the typical size of Ontario finishing barns (1000 to 2000 head). Pen selection was based on convenience sampling with regard to the ease of hanging ropes in high activity areas within pens. Pen occupancy rates varied from 15 to 200 pigs per pen, with most barns housing between 20 and 40 pigs per pen. Specific counts of pigs per pen or number of pigs interacting with a rope were not performed. Pigs were between 2 and 5 mo of age at the time of testing to ensure consistency with the common use of the oral fluids ELISA test in the field. All sampling was performed in finishing barns except for 2 farms on which late nursery pigs (2 mo of age) were tested. After permitting the pigs to chew the ropes for 20 to 30 min, ropes were collected from the pens and wrung individually into separate plastic bags. The fluid that pooled in each bag was then poured into 50 mL plastic centrifuge tubes (Fisherbrand; Fisher Scientific, Ottawa, Ontario) and the samples immediately transported at approximately 20°C to the Animal Health Laboratory at the University of Guelph where they were tested using a commercial PRRSV oral fluid antibody ELISA (IDEXX PRRS Oral Fluids Ab Test; IDEXX Laboratories, Westbrook, Maine, USA). Oral fluid ELISA test results were expressed as sample-to-positive (S/P) ratios. An S/P ratio ≥ 0.4 was considered positive according to the directions supplied by the manufacturer.

If oral fluid antibody testing results were not in agreement with the producer's assessment, serum samples (n = 10) were collected from pigs in the same pens where the oral fluids were collected within 1 wk of oral fluid sampling and then tested using a commercial PRRSV serum antibody ELISA (IDEXX PRRS X3 Ab Test). To collect serum samples, the animals were restrained, then bled from the jugular vein using an 18-gauge needle and a single use serum vacutainer tube (BD-Canada, Mississauga, Ontario). Serum samples were promptly submitted to the Animal Health Laboratory at the University of Guelph for antibody testing (IDEXX PRRS X3 Ab Test). Serum ELISA test results were likewise expressed as S/P ratios, with values ≥ 0.4 considered positive.

Three approaches for increasing the diagnostic specificity of the PRRS oral fluid ELISA were examined based on earlier reports from practitioners regarding sporadic false positive/low positive test results. These were:

i) Use individual pen S/Ps to establish barn status but raise the PRRSV oral fluid ELISA S/P cutoff to increase diagnostic specificity.
ii) Use the mean PRRSV ELISA S/P calculated from multiple pen oral fluid samples to establish barn status.
iii) Use serial testing to evaluate unexpected positive results.

Pigs in 2 herds were initially uninterested in the ropes but were coaxed into chewing by wetting the ropes slightly or rubbing them in the feed. One herd required sustained efforts, but 6 pen samples were collected after approximately 1 h. In the latter case, all 6 samples tested strongly positive (S/P range: 4.3 to 5.2; mean: 4.8), despite the initial reluctance of the pigs to chew on the rope and the small volume of oral fluid collected from each rope. In 2 of the study herds, the pigs pulled down some of the ropes which were not recovered. As a result, 4 samples were collected from 1 PRRSV naïve herd and 5 from another PRRSV naïve herd. Six samples were successfully collected from each of the remaining 29 herds for a total of 183 oral fluid samples.

Three herds initially reported as PRRSV naïve by the owner were reclassified as PRRSV positive based on strongly positive oral fluid antibody testing followed by strongly positive serum antibody testing. Thus, of the 31 participating herds, 23 were categorized as PRRSV naïve and 8 as PRRSV positive. This resulted in 135 oral fluid samples from naïve herds and 48 from PRRSV positive herds.

As shown in Figure 1A, oral fluid ELISA S/P values ranged from 0.0 to 0.44 in the 23 herds categorized as PRRSV negative and 0.1 to 7.4 in the 8 herds categorized as PRRSV antibody-positive. In herds classified as PRRSV negative, 2 of 135 oral fluid samples (1.5%) had S/P values at or above the 0.4 cutoff (0.43 and 0.44). These 2 samples originated from different herds. Both herds received pigs from PRRS naïve sources and continued to test PRRS negative on subsequent RT-PCR testing of oral fluids following the completion of this study. Among the PRRSV antibody-positive herds, 1 of 48 oral fluid samples (2.1%) had an S/P value below the 0.4 cutoff (0.1). As shown in Figure 1B, mean S/P values in negative herds ranged from 0.0 to 0.3. In contrast, mean S/P values in positive herds ranged from 2.4 to 6.2.

This study was undertaken to examine undocumented reports from the field stating that the oral fluid PRRS ELISA test produced an unacceptably high number of false positive results in PRRS naïve herds. The methodology applied here to investigate the reports of false positive results was limited due to the study's reliance on the owner's estimation of the PRRS status of

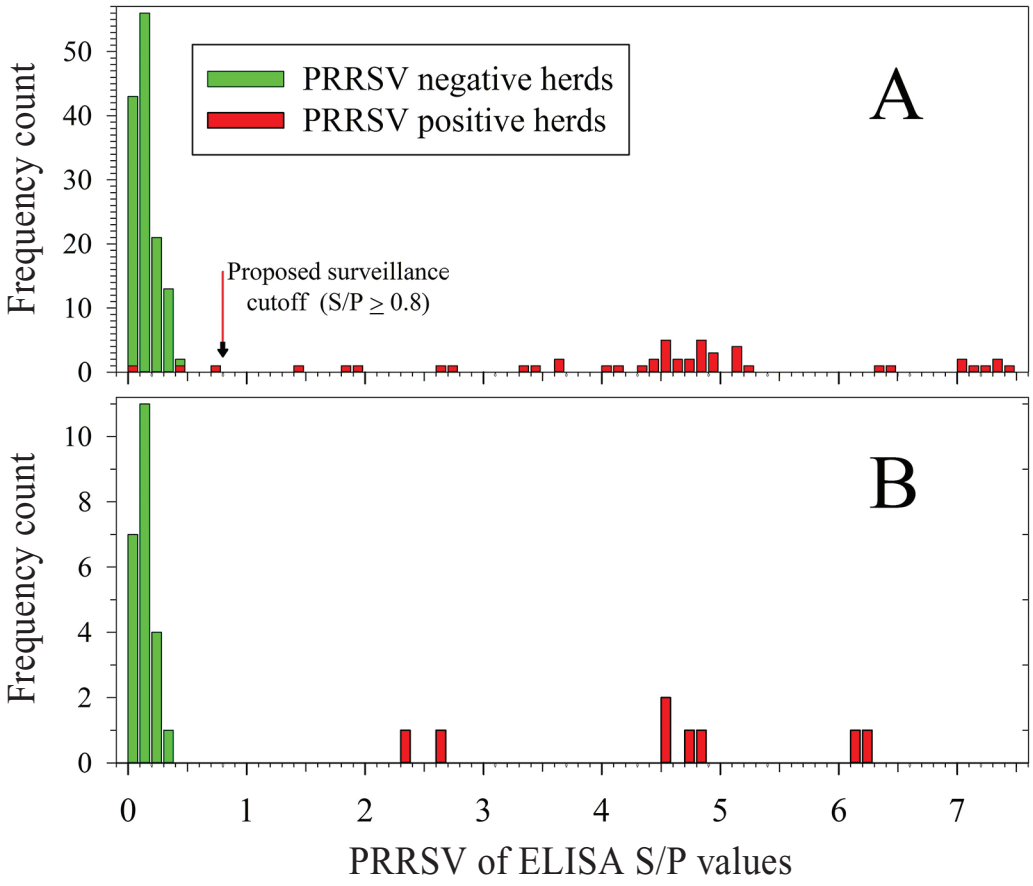


Figure 1. Frequency distribution of PRRSV oral fluid ELISA S/P test results (IDEXX PRRS Oral Fluids Ab Test; IDEXX Laboratories, Westbrook, Maine, USA). A – 135 oral fluid samples from PRRSV-negative herds ($n = 23$) and 48 oral fluid samples from PRRSV-positive herds ($n = 8$). B – Mean PRRSV oral fluid ELISA S/P calculated from the individual oral fluid test results (usually 6) from each herd.

the barns. It did, however, accurately reflect how the test was used by producers and practitioners, including the convenience sampling of pens within barns. The results of this study reflected the field reports with respect to unexpected positive results. The 1.5% false positive rate found in this study was consistent with the manufacturer's reported diagnostic specificity of 98.7% (Personal communication, Dr. Silvia Zimmerman, IDEXX, Westbrook, Maine, USA). Therefore, the methodology applied here likely identified the problems that practitioners were reporting regarding test accuracy, while being consistent with the manufacturer's stated value for test specificity. In certain situations, a 98.5% specificity is acceptable for diagnostic purposes. However, when used for regular disease surveillance purposes, a 1.5% rate of unexpected positive results ("false alarms") undermines confidence in the test (4). For example, if the oral fluids ELISA test was used routinely prior to moving pigs from a PRRS naïve nursery to a naïve finishing barn and 6 rope samples were submitted per group with 8 groups per year passing through the nursery, a false positive result could be expected on average to occur every 1 to 2 y. This would result in an unnecessary disruption of pig flow for the operators.

In this study, the PRRSV positive herds, individual pen sample S/Ps were as high as 7.5 and were generally much higher than the manufacturer's recommended cutoff of S/P $\geq$ 0.4 (5). Thus, 45 of the 48 pen samples from the 8 PRRSV antibody positive herds had S/P values $\geq$ 0.8. Three samples, each from a different PRRSV antibody positive herd, had S/P values of 0.1, 0.5, and 0.7. In fact, S/Ps $<$ 0.4 from a PRRSV positive barn may reflect the true PRRSV antibody status of that pen at the time of sampling because the virus can take weeks to spread to all pens of pigs within a facility. Regardless, if the 48 oral fluid samples from the 8 PRRSV positive herds in this study were assumed to originate from PRRS infected pigs, a cutoff of S/P $\geq$ 0.8 would result in a diagnostic sensitivity of 93.7% at the individual sample level. However, testing 6 samples per barn resulted in a diagnostic sensitivity of 100% at the barn level. Thus, setting the individual sample cut-off at S/P $>$ 0.8 resulted in 100% diagnostic specificity at the individual pen sample level as well as 100% sensitivity at the barn level when 6 samples were collected per barn. A more appropriate cut-point may be identified in the future when larger data sets are examined but using 0.8 in this study accurately categorized all the herds tested.

Porcine reproductive and respiratory syndrome virus infection may not be uniformly distributed throughout a positive barn. Collecting several oral fluid samples increases the probability of including a positive pen in the sampling, but larger sample sizes also increase the possibility of a false positive result. An approach that addresses the latter concern is to use the mean S/P value

calculated from several individual pen samples to establish barn status. In this study, the mean S/P calculated from 4 to 6 individual pen results clearly differentiated PRRSV negative (mean S/P range 0.0 to 0.3) from PRRSV antibody positive (mean S/P range: 2.4 to 6.2) barns and resulted in a diagnostic specificity of 100% at a cutoff of S/P $\geq$ 0.4. Using mean S/P values runs the risk of missing an early PRRSV infection by diluting a true low positive reactor that is captured on only 1 of several oral fluid samples, but this can be resolved by periodic testing over time.

The third approach to addressing unexpected low positive S/P ratios involves "serial testing" of the original sample using a test based on a different assay format, i.e., a different antibody test or a test for nucleic acid. For example, testing blood samples for PRRS antibodies as done in this trial can confirm or refute unexpected positive results from ELISA testing of oral fluids. Likewise, RT-PCR testing could be performed on an oral fluid sample with an S/P ratio $\geq$ 0.4 from a location where all other pen test results were $<$ 0.4. This would be particularly appropriate to exclude the possibility that a recent PRRSV introduction was the cause of a single low positive reaction. The latter approach does not require additional sample collection and could be requested as soon as the unexpected positive ELISA result was received. Serial testing is routinely used in disease control programs because false positive results invariably increase as disease prevalence declines.

In this study, oral fluid samples were found to be a practical and effective alternative to serum samples when testing for PRRSV antibodies. Collecting oral fluids using cotton ropes hung in pens was less stressful for pigs, as well as quick, simple, and convenient for animal caretakers. The results of this study (98.5% specificity) agreed with previous studies regarding the diagnostic specificity (98.7%) of the IDEXX PRRS oral fluids ELISA for individual samples. However, this level of diagnostic specificity at the individual pen sample level, is not sufficiently high for routine use in commercial swine herds. Three approaches to improving test specificity are proposed where multiple oral fluid samples are collected per location: i) use a positive cutoff of S/P $\geq$ 0.8 for individual pen oral fluid samples; ii) use the manufacturer's recommended cutoff of S/P $\geq$ 0.4, but establish barn status using the mean S/P calculated from several oral fluid samples (6 samples were used in this study); and iii) Use serial testing to confirm or refute unexpected oral fluid test results. RT-PCR testing is likely the most appropriate test when low positive reactors could result from an early stage PRRSV infection.

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